



ForgeX Introduction

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Motivation

Forged from the need for a highly scalable, cost-effective, and R&D-friendly power-electronics platform, the ForgeX family is designed to bridge the gap between experimental development and field-deployable hardware.

ForgeX is intended to remain accessible during development while providing the architectural discipline, maintainability, and electrical performance required for practical deployment.

The platform is designed not around a single application, but around a family of power-electronic systems that can evolve over time.

ForgeX treats experimentation, maintainability, and serviceability as architectural requirements rather than secondary considerations. Hardware is intended to remain accessible for probing, modification, replacement, and iterative development throughout its lifecycle, from initial research and prototyping through validation and field deployment.

Design Principles

ForgeX is guided by the following principles:

- **Modularity** — Functions should be independently replaceable where practical.
- **Electrical Locality** — Electrically sensitive and high-energy functions should remain physically localized.
- **Defined Interfaces** — Module boundaries should be explicit and documented.
- **Maintainability** — Hardware should remain accessible for inspection, troubleshooting, and replacement.
- **Incremental Evolution** — Improvements should not require unnecessary redesign of unrelated subsystems.
- **Performance Preservation** — Modularity should not be achieved at the expense of critical electrical performance.
- **Application Scalability** — The architecture should support a range of power-electronic applications and topologies.

Modular Architecture

ForgeX employs a highly modular hardware architecture in which major engineering functions are separated into independent, self-contained modules.

This modularity allows individual modules to be developed, tested, characterized, revised, and replaced independently, without requiring unnecessary changes to the remainder of the system. Where applicable, this separation can also simplify the evaluation of individual subsystems against relevant validation and certification requirements before system-level integration.

The objective is not modularity for its own sake. Module boundaries are chosen to preserve electrical locality, contain high-energy and high-frequency phenomena, and establish clear interfaces between engineering concerns.

ForgeX seeks to achieve modularity without compromising electrical locality. High-current paths, high- di/dt switching loops, gate-drive paths, local decoupling, high speed Digital Communication and other electrically sensitive structures are kept physically close to the functions they serve rather than being distributed merely for organizational convenience.

This allows improvements to be introduced incrementally while minimizing disruption to established portions of the system and preserving the electrical characteristics of critical subsystems.

Separation of Engineering Concerns

The ForgeX hardware architecture separates major engineering concerns into distinct functional domains, including:

- Processing and real-time control
- Industrial communication
- Power conversion and power switching
- Auxiliary low-voltage power
- Signal conditioning and measurement
- Power distribution and interconnection

These functions are connected through deliberately defined electrical and mechanical interfaces rather than being tightly coupled into a single monolithic control board.

This separation allows each domain to be developed, tested, and validated according to its own requirements while maintaining a coherent system architecture.

The separation also extends to development concerns. Changes to an encoder technology, communication interface, control processor, measurement subsystem, or power-stage implementation should not inherently require redesign of unrelated portions of the system.

This enables independent development and experimentation while maintaining well-defined boundaries between subsystems.

Interfaces and Compatibility

The interfaces between ForgeX modules form an important part of the architecture itself.

Interfaces are designed to establish stable electrical, mechanical, and functional boundaries between modules. Within a generation, these interfaces are intended to provide a defined compatibility baseline.

Across generations, compatibility is preserved selectively where the architectural and electrical requirements remain compatible.

A new generation is therefore not required to preserve every existing interface. Changes to interfaces should instead be deliberate, documented, and governed by defined compatibility requirements rather than occurring as an accidental consequence of implementation changes.

This allows ForgeX to evolve without sacrificing the benefits of a stable modular ecosystem.

Compatibility may apply independently to different aspects of an interface, including electrical, mechanical, functional, power, and communication requirements. This allows future generations to retain useful compatibility where practical while permitting architectural changes where they provide sufficient benefit.

Scalability and Evolution

The resulting architecture allows ForgeX to scale across a broad range of power-electronic applications.

A system may use only a subset of the available modules for a simple application, while more demanding systems may combine additional processing, sensing, communication, and power modules.

The same architectural principles can therefore be applied to different inverter and power-conversion topologies without requiring every application to begin from an entirely independent hardware design.

Although the initial ForgeX implementation is oriented toward motor-control and inverter applications, the architecture is not inherently restricted to a single application or topology. The same principles may be applied to applications such as VFDs, compressor drives, servo and actuator systems, renewable-energy inverters, and more complex multi-module or multilevel power converters where the required electrical and functional interfaces can be maintained.

ForgeX is consequently intended to be an evolving hardware family rather than a single fixed product.

Future generations may introduce new capabilities, revise existing modules, or replace individual implementations while preserving selected interfaces and architectural principles established by earlier generations.

ForgeX Module Semantics

ForgeX defines a set of functional module classes that describe the architectural role of a hardware module within the system.

These classifications are intended to remain independent of any particular implementation or generation. A module belonging to a given

class may therefore be redesigned, replaced, or extended in a future generation while retaining the same fundamental architectural role.

The base naming convention for ForgeX modules is:

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<Module Name>_G<Generation Number><Additional Specifiers>
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This convention identifies the module's architectural role and generation while allowing additional specifiers to distinguish implementation-specific characteristics or variants.

Example Module Names Include:

- HBM_G0VH4C5
- LVP_G0S4
- PMB_G0S4VH4
- PSU_G0P20N

HBM — Half-Bridge Module

The HBM provides a localized power-switching stage for the conversion system. It integrates the high-current switching elements with their associated gate-drive, measurement, and protection circuitry, providing a self-contained power-switching stage.

The HBM is intended to keep electrically sensitive and high- di/dt power-stage functions physically localized, while exposing defined interfaces to the remainder of the ForgeX system.

LVP — Low-Voltage Plane

The LVP provides the low-voltage control and signal domain of the system. It acts as an interface boundary between processing, measurement, sensing, and the power-stage hardware.

The LVP may contain signal conditioning, protection, low-voltage interfaces, sensing infrastructure, and other circuitry required to connect the control and processing domain to the wider system.

PMB — Power MotherBus

The PMB provides the physical and electrical interconnection between major power modules.

Its purpose is to distribute power and establish the required power distribution topology while keeping the high-current infrastructure centralized and mechanically organized.

The PMB is not intended to absorb the functions of the individual power modules; rather, it provides the infrastructure through which those modules are combined into a larger system.

Its implementation may range from a simple bus-bar-like structure providing only power distribution and interconnection, to a more integrated subsystem incorporating functions such as voltage and current measurement, protection, monitoring, or control.

PSU — Power Supply Module

The PSU provides auxiliary low-voltage power required by the ForgeX system and its modules.

The PSU is treated as an independent subsystem so that auxiliary power generation and regulation can evolve independently from the primary power-conversion and control hardware.

Processing / Development Modules

Processing modules provide the computational and real-time control functions of the system.

These modules may host the primary control processor, firmware interfaces, development infrastructure, or other computational functions while remaining separated from the power-stage hardware.

Interface Modules

Interface modules provide specialized connections between ForgeX and external devices or subsystems.

These may include encoder interfaces, analog measurement, communication, isolation, or other signal-conditioning functions.

The module classification describes the architectural responsibility of a module, rather than prescribing a fixed PCB boundary. A future generation may combine, split, or reorganize these functions where doing so provides a meaningful architectural advantage while preserving the relevant ForgeX interfaces.

For example, the Gen0 LVP integrates BISS-C and quadrature encoder interfaces onboard while exposing a simple digital interface to the processing domain. Future generations may instead implement these functions as separate modules or dedicated interface devices where doing so provides an architectural advantage.